

4th week

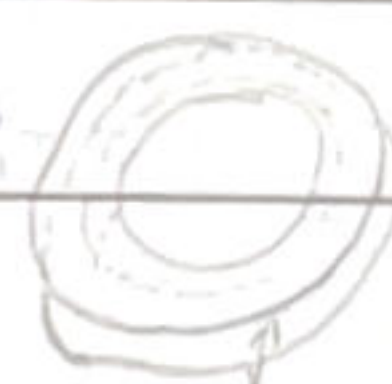
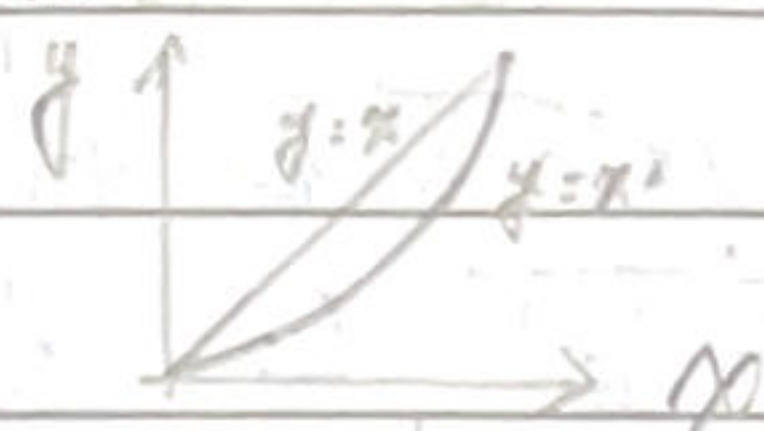
r: 高点的水平距离

1. Region Bounded by
- $y=x^2$
- and
- $y=x$
- about the
- y
- axis.

The Volume is

$$V = \int_0^1 2\pi x(x-x^2) dx = 2\pi \int_0^1 (x^2-x^3) dx$$

$$= 2\pi \left(\frac{1}{3}x^3 - \frac{1}{4}x^4 \right) \Big|_0^1 = \frac{\pi}{6}$$



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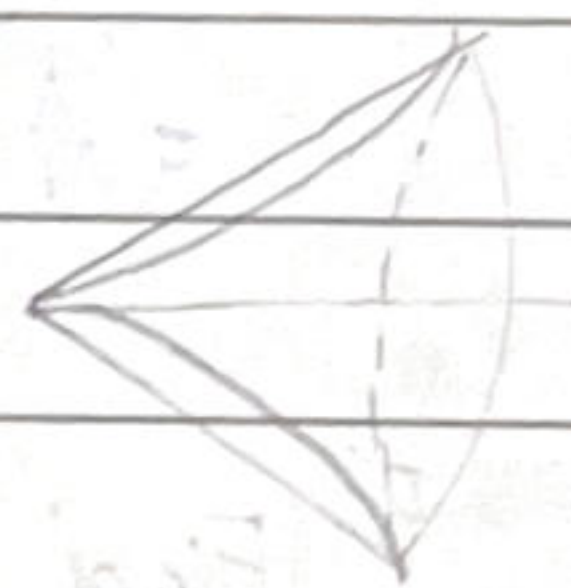
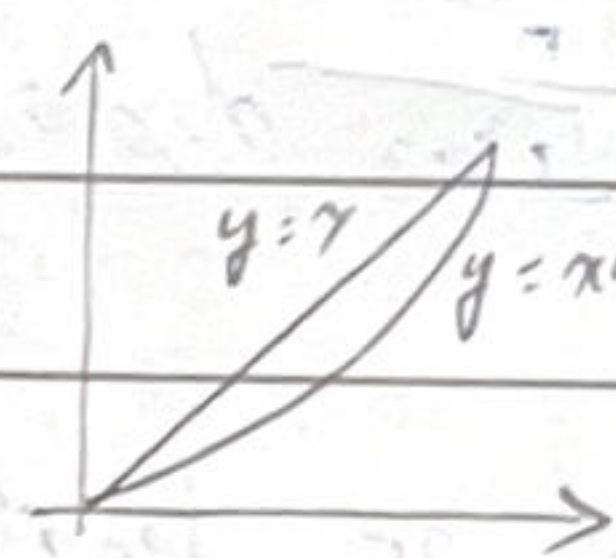
$$= 2\pi \left(\frac{1}{3}x^3 - \frac{1}{4}x^4 \right) \Big|_0^1 = \frac{\pi}{6}$$

2. Region Bounded by
- $y=x^2$
- and
- $y=x$
- about the
- x
- axis.

The Volume is

$$V = \int_0^1 2\pi (y)(\sqrt{y}-y) dy = 2\pi \int_0^1 (y^{\frac{3}{2}}-y^2) dy$$

$$= 2\pi \left(\frac{2}{5}y^{\frac{5}{2}} - \frac{1}{3}y^3 \right) \Big|_0^1 = 2\pi \left(\frac{1}{15} \right) = \frac{2\pi}{15}$$



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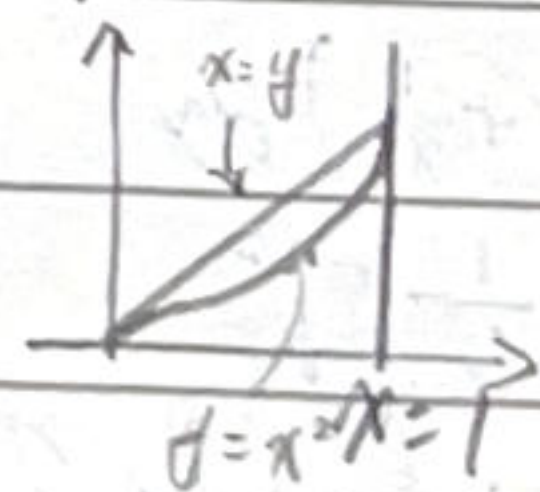
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3. Region Bounded by
- $y=x^2$
- and
- $y=x$
- about the line
- $x=1$

The Volume is

$$V = \int_0^1 2\pi (1-x)(x-x^2) dx = 2\pi \int_0^1 (x^3-2x^2+x) dx$$

$$= 2\pi \left(\frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 \right) \Big|_0^1 = \pi \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{2} \right) = \frac{\pi}{6}$$



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4th week ARC LENGTH $L = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} = \sqrt{\Delta x_i^2 + \Delta y_i^2} = \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} \cdot \Delta x_i = \sum_{i=1}^n \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} \Delta x_i = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

4 Show that the perimeter of a circle of radius r is $2\pi r$.

let $y = \sqrt{r^2 - x^2}$ be the function of the semi-circle

$$\begin{aligned} L &= 4 \int_0^r \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 4 \int_0^r \sqrt{1 + \left(\frac{-x}{\sqrt{r^2 - x^2}}\right)^2} dx \\ &= 4 \int_0^r \sqrt{1 + \frac{x^2}{r^2 - x^2}} dx = 4 \int_0^r \sqrt{\frac{r^2}{r^2 - x^2}} dx = 4 \int_0^r \frac{r}{\sqrt{r^2 - x^2}} dx \\ &= 4r \left(\sin^{-1} \left(\frac{x}{r} \right) \right) \Big|_0^r = 4r (\sin^{-1} 1 - \sin^{-1} 0) = 2\pi r \end{aligned}$$

let $y = \sqrt{r^2 - x^2}$ be the function of the semi-circle

$$\begin{aligned} L &= 4 \int_0^r \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= 4 \int_0^r \sqrt{1 + \left(\frac{-x}{\sqrt{r^2 - x^2}}\right)^2} dx \\ &= 4 \int_0^r \frac{r}{\sqrt{r^2 - x^2}} dx = 4r (\sin^{-1} \frac{x}{r}) \Big|_0^r = 2r (\sin^{-1} 1 - \sin^{-1} 0) = 2\pi r \end{aligned}$$

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5 Find the arc length of the curve $y = \frac{x^3}{6} + \frac{1}{2x}$, $x \in [1, 2]$

$$y = \frac{x^3}{6} + \frac{1}{2x} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^2 - \frac{1}{2x^2}$$

$$L = \int_1^2 \sqrt{1 + \left(\frac{1}{2}x^2 - \frac{1}{2x^2}\right)^2} dx$$

$$= \int_1^2 \sqrt{1 + \frac{1}{4}\left(x^2 - \frac{1}{x^2}\right)^2} dx$$

$$= \int_1^2 \sqrt{1 + \frac{1}{4}\left(x^4 - 2 + \frac{1}{x^4}\right)} dx$$

$$= \int_1^2 \sqrt{\frac{1}{4}x^4 + \frac{1}{2} + \frac{1}{4x^4}} dx$$

$$= \int_1^2 \frac{1}{2} \sqrt{x^4 + 2 + \frac{1}{x^4}} dx$$

$$= \frac{1}{2} \int_1^2 \left(x^2 + \frac{1}{x^2}\right) dx$$

$$= \frac{1}{2} \left(\frac{1}{3}x^3 - \frac{1}{x}\right) \Big|_1^2$$

$$= \frac{1}{2} \left(\frac{8}{3} - \frac{1}{2} - \frac{1}{3} + 1\right) = \frac{17}{12}$$

$$\frac{1}{3} - \frac{1}{2}$$

$$\frac{21}{6} - \frac{2}{6}$$

$$\frac{19}{6}$$

$$\frac{1}{4}$$

$$y = \frac{x^3}{6} + \frac{1}{2x} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^2 - \frac{1}{2x^2}$$

$$L = \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

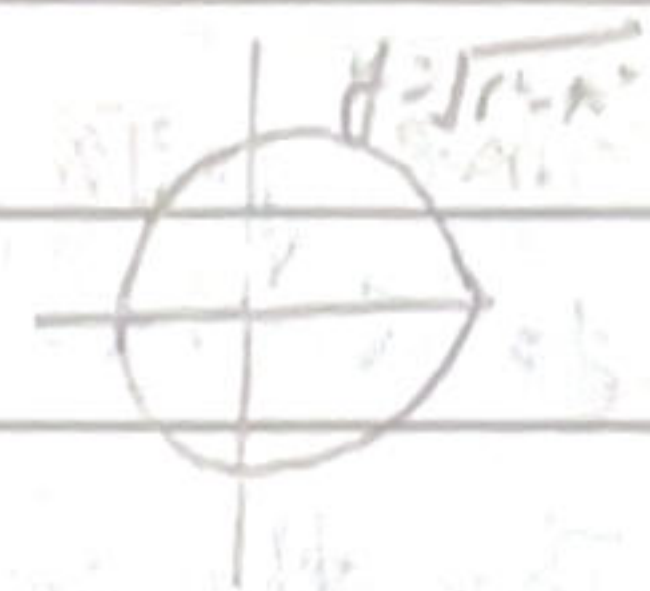
$$= \int_1^2 \sqrt{1 + \frac{1}{4}\left(x^4 + \frac{1}{x^4} - 2\right)} dx$$

$$= \int_1^2 \frac{1}{2} \sqrt{x^4 + 2 + \frac{1}{x^4}} dx$$

$$= \frac{1}{2} \int_1^2 \left(x^2 + \frac{1}{x^2}\right) dx$$

$$= \frac{1}{2} \left(\frac{1}{3}x^3 - \frac{1}{x}\right) \Big|_1^2$$

$$= \frac{1}{2} \left(\frac{8}{3} - \frac{1}{2} - \frac{8}{3} + 1\right) = \frac{17}{12}$$

4th week6 Show that the surface area of a sphere of radius r is $4\pi r^2$.let $y = \sqrt{r^2 - x^2}$ be the function of the upper semi-circle of the radius r .

Then the surface area of the sphere is

$$\begin{aligned}
 S &= \int_{-r}^r 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx &= 4\pi r \int_0^r \sqrt{\frac{r^2 - x^2}{r^2 - x^2}} dx \\
 &= 4\pi \int_0^r y \sqrt{1 + \left(\frac{-2x}{2\sqrt{r^2 - x^2}}\right)^2} dx &= 4\pi r (x) \Big|_0^r = 4\pi r^2 \\
 &= 4\pi \int_0^r y \sqrt{\frac{r^2 - x^2 + x^2}{r^2 - x^2}} dx
 \end{aligned}$$

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 &= 4\pi \int_0^r y \sqrt{1 + \left(\frac{-x}{\sqrt{r^2 - x^2}}\right)^2} dx \\
 &= 4\pi \int_0^r y \sqrt{\frac{r^2 - x^2 + x^2}{r^2 - x^2}} dx \\
 &= 4\pi \int_0^r \frac{r^2 - x^2}{r^2 - x^2} dx \\
 &= 4\pi r (x) \Big|_0^r = 4\pi r^2
 \end{aligned}$$

let $y = \sqrt{r^2 - x^2}$ be the function of the upper semi-circle of the radius r .

Then the surface area of the sphere is

$$\begin{aligned}
 S &= \int_{-r}^r 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\
 &= 4\pi \int_0^r \sqrt{r^2 - x^2} \left(\sqrt{1 + \left(\frac{-x}{\sqrt{r^2 - x^2}}\right)^2} \right) dx \\
 &= 4\pi \int_0^r \frac{\sqrt{r^2 - x^2} \sqrt{r^2 - x^2 + x^2}}{\sqrt{r^2 - x^2}} dx \\
 &= 4\pi r \int_0^r dx \\
 &= 4\pi r (x) \Big|_0^r \\
 &= 4\pi r^2
 \end{aligned}$$

4th week

7. Find the area of the surface obtained by rotating the curve $y = \sqrt[3]{x}$, $0 \leq x \leq 1$ about the y-axis.

$$y = \sqrt[3]{x} \Rightarrow x = y^3 \Rightarrow \frac{dx}{dy} = 3y^2$$

Then the surface area is

$$S = \int_0^1 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= 2\pi \int_0^1 y^3 \sqrt{1 + (3y^2)^2} dy$$

$$= 2\pi \int_0^1 y^3 \sqrt{1 + 9y^4} dy$$

$$\text{let } u = 1 + 9y^4 \Rightarrow du = 36y^3 dy$$

$$S = \frac{1}{18} \pi \int_1^{10} \sqrt{u} du$$

$$= \frac{\pi}{18} \cdot \frac{2}{3} (u^{\frac{3}{2}}) \Big|_1^{10}$$

$$= \frac{\pi}{27} (\sqrt{1000} - 1)$$

$$= \frac{\pi}{27} (10\sqrt{10} - 1)$$

$$y = \sqrt[3]{x} \Rightarrow x = y^3 \Rightarrow \frac{dx}{dy} = 3y^2$$

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